# ESLtopia — Test Scenario Coverage Checklist

Every scenario that should be covered, and its current automation status · v1.6 · 2026-07-06

| | |
|---|---|
| Purpose | A complete inventory of test scenarios across the application — not just what's automated, so gaps are visible and can be prioritized deliberately. |
| Companion documents | Master Test Plan · Test Case Specification (detail for every AUTOMATED/PARTIAL item below) |
| Legend | AUTOMATED covered by the current suite PARTIAL some coverage, notable gap remains GAP not automated today |

**v1.6 revision:** §10 (Cross-Cutting): cross-browser coverage flips from GAP to AUTOMATED. Two new `playwright.config.ts` projects run the existing UI/API suite under Firefox and WebKit (pure-REST files and the visual suite excluded — see Test Plan §13). No new test IDs were created; this is the same 144 cases now executed under up to three engines. Verified clean: the one failure and one flake seen in an initial full cross-browser run were confirmed by isolated reruns to be pre-existing parallel-load flakiness, not real per-browser defects.

**v1.5 revision:** Closed the three high-priority items from `test-coverage-todo.md`. §6 (Classes Management): all three Classes-tab UI click-through rows flip from GAP to AUTOMATED (CLSUI-01–04). §10 (Cross-Cutting): the XSS/HTML-injection row flips from GAP to AUTOMATED (CLSUI-05/06, RECUI-09). §1 (Authentication): the rate-limiting row flips from GAP to **PARTIAL**, not AUTOMATED — LOGIN-17 and FPW-08 now exercise a burst of repeated attempts and lock in the invariant that failed attempts never succeed and the server never 500s, but as observed 2026-07-06 no lockout or 429 occurs within that burst on either endpoint, so the underlying protection this scenario describes isn't actually confirmed to exist. See the Test Plan §11 risk note.

**v1.4 revision:** A doc-sync pass against the actual `tests/` directory found three things this checklist had missed: (1) the visual-regression suite (§10) was live in CI but unlisted here; (2) a signup anti-enumeration test (REG-11) existed in code but wasn't reflected in §1; (3) an RLS insert-forgery test (CLS-08, renumbered from further down the file) existed in code but wasn't reflected in §3/§6. All three are now AUTOMATED rows below. No new test-writing happened in this pass — this only corrects the checklist to match code that already existed.

**v1.3 revision:** Record-sync failure handling (§7) is now closed — RECUI-08 simulates a failed save (a resolved Supabase-error response, not a network exception) and confirms both that the error is logged and that the write still doesn't persist, since logging isn't durability. Both scenario gaps the v1.1 code review surfaced are now automated.

**v1.2 revision:** The teacher-removal cascade/warning gap is now closed — SCHT-14 and SCHT-15 (§5) verify the cascade actually deletes a populated teacher's classes/records and that the confirm dialog discloses it. Record-sync failure handling (§7) remains an open gap.

**v1.1 revision:** A code review pass fixed 8 code-level issues (see the Test Case Specification's revision note) but did not add new automated tests, so it surfaced — rather than closed — two scenario gaps: the teacher-removal cascade/warning behavior (§5) and record-sync failure handling (§7).

## 1. Authentication & Account Lifecycle

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Valid login logs the user in | LOGIN-01, LOGIN-07/08/09 |

| | | |
|---|---|---|
| AUTOMATED | Wrong password / unregistered email → generic, non-enumerating error | LOGIN-02/03/10/11 |
| AUTOMATED | Client-side validation: invalid email format, empty email/password | LOGIN-04/05/06 |
| AUTOMATED | Unconfirmed user cannot log in, with the same generic message | LOGIN-16 |
| AUTOMATED | Signup happy path (teacher and school) through confirmation to logged-in | REG-01/02 |
| AUTOMATED | Signup form validation: mismatched/short/missing fields | REG-03–06 |
| AUTOMATED | Signup role is restricted server-side even if the client is bypassed | REG-10, ROLE-02 |
| AUTOMATED | Signup with an already-registered email does not reveal that an account already exists (anti-enumeration) | REG-11 — existed in code, added to this checklist 2026-07-06 |
| AUTOMATED | Weak password rejected on both signup and reset | REG-12, PWR-03/04 |
| AUTOMATED | Forgot-password flow does not reveal whether an email is registered | FPW-01, FPW-04/05 |
| AUTOMATED | Completing a password reset invalidates the old password and the link itself | PWR-01, PWR-05 |
| AUTOMATED | Sign out ends the session and does not silently restore it on reload | LOGOUT-01/02 |
| AUTOMATED | Pending / inactive accounts are blocked at login with a specific message | ROLE-04/05, SCHT-06 |
| PARTIAL | Email deliverability and content (confirmation, invite, reset mail) | Confirmation is simulated via the admin API, not by reading a real inbox — deliberate tradeoff (see Test Plan §11), but actual email rendering/copy is unverified. |
| PARTIAL | Rate limiting / brute-force protection on login and password-recovery endpoints | LOGIN-17 and FPW-08 now exercise a small burst (4–5) of repeated attempts and confirm the invariant that failed attempts never succeed and the server never 500s. But as observed 2026-07-06, no lockout or 429 occurred within that burst on either endpoint — the test documents current behavior, it does not confirm brute-force protection actually exists. See Test Plan §11. |
| GAP | Account lockout / suspicious-activity handling | No such feature exists in the app today; flag if added. |

## 2. Session Management

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Unauthenticated visit shows the login screen, not the app shell | NAV-01 |
| AUTOMATED | A valid session persists across a full page reload | NAV-02 |

| | | |
|---|---|---|
| AUTOMATED | A session for a since-deactivated account is force-signed-out on reload | NAV-03 |
| AUTOMATED | A revoked/expired refresh token cannot mint new access tokens | LOGIN-15, LOGOUT-04 |
| PARTIAL | Access-token expiry mid-session (not just at the refresh-endpoint level) | Only the raw refresh-grant API path is tested; a real in-browser session sitting past token expiry while mid-action isn't simulated. |
| GAP | Multiple concurrent sessions/tabs for the same user | Not exercised. |

## 3. Authorization, Roles & RLS

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Each role (teacher/school/superadmin) sees only its intended tabs | ROLE-03, SCHT-05, SADM-01 |
| AUTOMATED | A user can never elevate their own role via direct REST calls | ROLE-11/12, PROF-03 |
| AUTOMATED | A user cannot read, modify, or forge ownership of another user's profile, class, or records | PROF-04, SADM-08/10, CLS-07/08/09/13 |
| AUTOMATED | Superadmin can read/update/delete across all users' data; regular roles cannot | SADM-04–15, CLS-15/16 |
| AUTOMATED | JWT claims (role, email, sub, exp) are well-formed for every role | ROLE-06–09, ROLE-13 |
| PARTIAL | School-scoped data isolation (school A cannot see school B's teachers) | SCHT-11 covers this at the API level; gated behind the undeployed feature flag. |
| GAP | Row-level security across every future table added to the schema | Structural risk, not a specific gap — flag whenever a new table is introduced without an accompanying RLS test. |

## 4. Superadmin — Admin Panel

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Admin tab visible only to superadmin; user table renders with Delete buttons | SADM-01/02/03 |
| AUTOMATED | Changing a user's role via the Admin panel dropdown persists to the DB | SADM-16 |
| GAP | Deleting a user via the Admin panel's Delete button (UI click-through) | Deletion is only tested at the REST layer (SADM-12); no test clicks the actual Delete button and confirms the row disappears from the table. |
| GAP | Admin panel search/filter/pagination, if the user table grows large | Not applicable today at current user counts; revisit if the panel gains these controls. |

## 5. School → Teacher Management Feature not yet deployed

| Status | Scenario | Notes |
|---|---|---|

| | | |
|---|---|---|
| AUTOMATED | School can invite (create), deactivate/reactivate, and remove a teacher | SCHT-01, 04, 07 — automatic once the flag is on |
| AUTOMATED | Cross-school authorization is enforced on every management action | SCHT-02, 03, 11 |
| AUTOMATED | Pending/deactivated teachers cannot log in; can self-activate via RPC | SCHT-06, 10, 12 |
| AUTOMATED | Edge function input validation (missing email, bad action, duplicate email) | SCHT-08, 09, 13 |
| GAP | Employees panel UI click-through (invite form, deactivate button, roster list) | Only tab visibility (SCHT-05) is UI-driven; the actual invite/manage UI has no click-through test — everything else calls the edge functions directly. |
| GAP | Invited-teacher email content and the setup link (?setup=1) flow end-to-end | Not exercised via UI. |
| AUTOMATED | Removing a teacher actually cascades to delete their classes/records, and the confirm dialog accurately warns about it | SCHT-14, SCHT-15 — added 2026-07-03 to close a gap a code review surfaced: SCHT-07's "remove" happy-path test previously gave the teacher no classes/records to lose, so the cascade (classes.owner_id / records.owner_id both cascade from auth.users) was never actually exercised. |

## 6. Classes Management

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Class CRUD (create, list, rename, delete) at the data layer | CLS-01–04 |
| AUTOMATED | Adding/removing students from a class's roster | CLS-05/06 |
| AUTOMATED | Data isolation between teachers on read/delete; superadmin can see everything | CLS-07/09, 15/16 |
| AUTOMATED | A teacher cannot forge another teacher's owner_id to insert a class under someone else's ownership | CLS-08 — existed in code, added to this checklist 2026-07-06; closes the write-side half of the isolation story that CLS-07/09 (read/delete) already covered |
| AUTOMATED | Deleting a class cascades to delete its records | CLS-14 |
| AUTOMATED | Classes tab UI click-through: creating a class via the "add class" form | CLSUI-01 |
| AUTOMATED | Classes tab UI click-through: adding/removing a student via the roster input | CLSUI-02/03 |
| AUTOMATED | Classes tab UI click-through: deleting a class via its delete button | CLSUI-04 |
| GAP | Duplicate student names within a class; very long/unicode class or student names | No boundary/edge-case input testing on these free-text fields. |

## 7. Records — Attendance, Grades & Payment

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Empty state when a teacher has no classes yet | RECUI-01 |

| | | |
|---|---|---|
| AUTOMATED | Selecting a class populates the student table | RECUI-02 |
| AUTOMATED | Marking a student present/absent updates the row and the daily summary | RECUI-03 |
| AUTOMATED | "All present" quick-fill action | RECUI-04 |
| AUTOMATED | Setting a grade and toggling payment persist across a reload (debounced autosave) | RECUI-05/06 |
| AUTOMATED | Records CRUD and cross-teacher isolation at the data layer | CLS-10–13 |
| PARTIAL | Student profile modal: attendance stats and note persistence | RECUI-07 covers the top-level stats and notes field; the trimester breakdown (attendance/grade history per term, final-grade dropdown) is not covered — see below. |
| GAP | Trimester breakdown in the student profile: expanding a term, viewing its attendance/grade log, setting a final grade | Rich sub-feature of the profile modal with no test coverage at all. |
| GAP | Changing the "class date" / "month" pickers to view a different day/month's data | Every current test uses "today"/"this month" implicitly; switching dates is untested. |
| GAP | Concurrent edits to the same record from two sessions (last-write-wins behavior) | The debounced upsert has no defined conflict resolution test. |
| AUTOMATED | A failed record save is logged, and does not silently count as persisted | RECUI-08 — added 2026-07-03. Intercepts the debounced upsert with a simulated 500 (a resolved error response, not a thrown network exception, to exercise the actual if (error) branch) and confirms both that the console error fires and that a reload shows the change was never actually saved. |

## 8. Worksheet Generator

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Selecting a topic and generating a worksheet opens the PDF preview with correct topic/grade metadata | PDF-01 |
| PARTIAL | Worksheet generation across all task types (match, true/false, fill-in, listen-and-circle, color-the-boxes) | Only the fillin type is exercised (via the "am / is / are" topic, chosen specifically because it's deterministic). The other four task-type renderers have no generation test. |
| GAP | Grade filter (dropdown that narrows the topic grid to one grade) | Not exercised — the suite always leaves "All grades" selected. |
| GAP | Exercise-type pills (Match vs. True/False) for topics that support both | Not exercised. |
| GAP | Exercise-count input bounds (min 5 / max 20, non-numeric input handling) | Not exercised. |
| GAP | Assigning a class/student to a worksheet and confirming Name/Class fields populate on the sheet | Current PDF-modal tests generate without selecting a class or student. |

| | | |
|---|---|---|
| GAP | Background worksheet toolbar (the "Show answers" / "New set" / "Preview PDF" controls behind the modal, before it's opened) | Generation immediately opens the modal in every existing test; the pre-modal toolbar path is unexercised. |

## 9. PDF Preview Modal

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Answer-key toggle shows/hides answers correctly | PDF-02 |
| AUTOMATED | "New set" regenerates content while keeping the modal open | PDF-03 |
| AUTOMATED | Modal closes via its close button and via the Escape key | PDF-04/05 |
| GAP | "Print" button triggers window.print() with the expected print-only CSS applied | Click handler exists; actual browser print behavior and the @media print rules are not verifiable via Playwright's normal DOM assertions. |
| GAP | "Download PDF" opens a new tab with the worksheet HTML and expected styling | Not exercised — involves a new-window/popup flow and print-triggered download that's awkward to assert headlessly. |
| GAP | Pop-up-blocked fallback (alert shown when window.open returns null) | Not exercised. |
| GAP | Modal scroll position resets to top when "New set" is clicked | Behavior exists in code (pdfScrollRef.scrollTo) but is unverified. |

## 10. Cross-Cutting Concerns

| Status | Scenario | Notes |
|---|---|---|
| AUTOMATED | Visual/pixel-diff regression on four key screens (login, dashboard, worksheet generator settings, PDF preview modal) at a fixed 1280×900 viewport | VIS-01–04, `tests/visual/visual-regression.spec.ts` — existed in code (merged via the visual-regression-tests branch) but was missing from this checklist until now. Baselines are generated on CI's ubuntu-latest runner; regenerate via the "Update Visual Regression Baselines" workflow after intentional UI changes. |
| AUTOMATED | Cross-browser coverage (Firefox, WebKit — desktop) | Same 144 cases as elsewhere in this document, now also executed under the `firefox` / `webkit` Playwright projects (minus the pure-REST and visual-regression files — see Test Plan §13). Mobile browser emulation remains a GAP. |
| GAP | Responsive/mobile layout correctness at the 860px and 480px breakpoints | CSS exists for both; no viewport-sized visual assertions. Distinct from the new visual-regression suite above, which only covers one fixed desktop viewport (1280×900), not these breakpoints. |
| GAP | Accessibility: screen-reader labeling, full keyboard navigation, color contrast | Only incidental keyboard support (Escape-to-close) is covered. |
| GAP | Performance under load (many classes/students/records for one teacher) | No defined SLA for this ~10-user portfolio app; out of scope per Test Plan. |
| AUTOMATED | XSS/HTML-injection resilience in free-text fields (student names, class names, notes) | CLSUI-05 (class name), CLSUI-06 (student name), RECUI-09 (notes) — confirm React's default escaping holds by injecting `<script>` / `<img onerror>` payloads and checking they render as literal text with no matching DOM element and no execution. |
| GAP | Network-failure handling (Supabase request timeouts, offline state) | No offline/error-banner UX exists to test today. |

| | | |
|---|---|---|
| PARTIAL | Deploy verification (confirming a shipped change is actually live in production) | Done manually per Test Plan §11 (bundle-hash check before trusting prod-run results) — not itself an automated Playwright test. |

## 11. Summary

Of the areas surveyed above, the following are the highest-value gaps to close next, in rough priority order:

1. Worksheet generation across the remaining four task types (match, true/false, listen-and-circle, color-the-boxes), not just fill-in.
2. Student profile trimester breakdown (attendance/grade history per term, final-grade entry).
3. Employees panel UI click-through, once the school-teachers feature is deployed and the edge-function tests are unblocked.
4. Generator settings (grade filter, exercise-type pills, exercise-count input bounds) and the remaining Medium-priority items in `test-coverage-todo.md`.

**Resolved 2026-07-06 (cross-browser):** Firefox and WebKit coverage (§10) is now automated via two new `playwright.config.ts` projects running the existing suite (minus pure-REST files and visual regression). No new test IDs; verified clean against production with no genuine per-browser defects found.

**Resolved 2026-07-06 (test-writing):** all three former High-priority backlog items are now automated or documented: Classes tab UI click-through (§6, CLSUI-01–04), XSS/HTML-injection resilience (§10, CLSUI-05/06 + RECUI-09), and rate-limiting behavior on login/recover (§1, LOGIN-17/FPW-08 — flipped to PARTIAL rather than AUTOMATED, since the tests document that no such protection is currently observed rather than confirming one exists). See the Test Plan §11 risk note on that last point.

**Resolved 2026-07-06 (doc sync):** this checklist was also brought back in sync with the code — the visual-regression suite (§10, VIS-01–04) is now listed as AUTOMATED, and two test cases that existed in code but were undocumented (REG-11 anti-enumeration in §1, CLS-08 RLS-insert-forgery in §3/§6) are now reflected here.

**Resolved 2026-07-03:** both scenario gaps the v1.1 code review surfaced — teacher-removal cascade (§5, SCHT-14/15) and record-sync failure handling (§7, RECUI-08) — are now automated.

ESLtopia Test Scenario Coverage Checklist · v1.6 · Generated 2026-07-06